Output Ontology (OO)

Score: 1/5 — Serious concern | Confidence: high

***Benchmark:** LAILA* | ***Context:** Arab High-School Arabic Essay Feedback (Qatar Primary)*

Output Ontology definition: Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.

> Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

This is the most severe misalignment. LAILA's output ontology is exclusively numeric ordinal trait scoring on the CAST rubric: 6 traits at 0–5 plus REL at 0–2 [Q34, Q36], with holistic computed as the sum [Q35, Q182]. All benchmarked models 'followed a multitask setup predicting holistic and trait scores jointly' [Q48]; LLMs produce trait scores in JSON format [Q181]; AraT5 formulates AES as text generation of trait names plus scores [Q147]. The deployment requires actionable natural-language revision suggestions explaining why points were lost [elicitation A3, regional YAML output_function], a qualitatively different output category that is entirely absent from the benchmark's output schema. Web research confirms this gap is field-wide: 'crucial aspects of writing evaluation, such as providing feedback, have often been overlooked' even in English [WEB-12], and no Arabic rubric-to-natural-language feedback system exists [WEB-13, WEB-23]. The CAST rubric is also explicitly genre-scoped to persuasive/argumentative writing [Q118], compounding the misalignment with deployment genres. The narrow REL scale (0–2) further limits granularity [Q91]. OO is HIGH priority and the mismatch is direct, structural, and acknowledged by the authors who 'explicitly prohibit its use for high-stakes educational decisions' [Q104].

ID	Evidence
Q34	"The rubric covers 7 writing traits [D4]: Relevance (REL), Organization (ORG), Vocabulary (VOC), Style (STY), Development (DEV), Mechanics (MEC), and Grammar (GRA)." (p.5)
Q35	"Additionally, a Holistic score (HOL) was computed as the sum of the 7 trait scores." (p.5)
Q36	"Six traits (all but REL) were rated on a 6-point scale (0 = lowest, 5 = highest), while REL was rated on a 3-point scale (0 = not relevant, 1 = partially relevant, and 2 = fully relevant)." (p.5)
Q48	"The models fall into three categories: feature-based (FB), encoder-based, and large language models (LLMs). LLMs were evaluated in zero-shot and few-shot settings, and all models followed a multitask setup predicting holistic and trait scores jointly." (p.7)
Q91	"Fourth, the relevance trait is scored on a narrow scale from 0 to 2, which limits the granularity of evaluation and may cause QWK to underrepresent subtle differences in model predictions." (p.10)
Q104	"We explicitly prohibit its use for high-stakes educational decisions affecting individual students." (p.10)
Q118	"Table 6: CAST Persuasive/Argumentative Writing Rubric - English Translation (Bashendy et al., 2024)." (p.13)
Q147	"The AES task is formulated as a text generation problem, where the model predicts the trait scores sequentially." (p.16)
Q181	"In the zero-shot setting, the LLM is tasked with generating trait scores in JSON format, given the prompt text, essay, and trait rubrics." (p.18)
Q182	"The holistic score was computed as the sum of the individual trait scores." (p.18)
WEB-12	AES literature explicitly identifies feedback generation as an overlooked area (https://link.springer.com/article/10.1007/s10462-024-11017-5)
WEB-13	AAEE Saudi system provides corrected essays but not trait-grounded revision suggestions (https://www.researchgate.net/publication/343796441_Automated_students_arabic_essay_scoring_using_trained_neural_network_by_e-jaya_optimization_to_support_personalized_system_of_instruction)
WEB-23	No Arabic system bridges trait scores to actionable natural-language feedback for secondary students (https://www.sciencedirect.com/science/article/abs/pii/S0306457319304935)

Strengths

- Trait-decomposed scoring offers a principled construct decomposition (REL, ORG, VOC, STY, DEV, MEC, GRA) that could anchor downstream feedback generation [Q34]
- CAST rubric is institutionally grounded in QUTC and provided in Arabic [Q33, Q114]
- Zero-rule for irrelevant essays (REL=0 → all traits 0) reflects pedagogical logic [Q37, Q117]

ID	Evidence
Q33	"All essays in LAILA dataset were scored using the Core Academic Skills Test (CAST) rubric [D7], developed by the Qatar University Testing Center (QUTC)." (p.5)
Q34	"The rubric covers 7 writing traits [D4]: Relevance (REL), Organization (ORG), Vocabulary (VOC), Style (STY), Development (DEV), Mechanics (MEC), and Grammar (GRA)." (p.5)
Q37	"Furthermore, if an essay received a REL score of 0, all other trait scores were set to 0, as irrelevant responses are not subject to further evaluation." (p.5)
Q114	"For annotating LAILA, we adopted the rubric from the Core Academic Skills Test (CAST), which is provided in Arabic, developed by the Qatar University Testing Center (QUTC)." (p.12)
Q117	"Note: A score of 0 is assigned if the student does not attempt the task, provides a response that falls below the performance level described for score 1, or submits content that is not relevant to the topic of the given prompt." (p.13)

Checklist

Checklist item definitions:

ID	Assessment Question
OO-1	Review output label categories for regional relevance
OO-2	Identify missing regionally specific categories
OO-3	Flag categories encoding non-regional values
OO-4	Consider stakeholder-driven taxonomy redesign
OO-5	Document taxonomy issues harming structural/content validity

Checklist responses:

OO-1: The 7-trait label space is broadly relevant to Arab high-school writing assessment, but the rubric is explicitly designed for persuasive/argumentative writing [Q118], limiting regional relevance for literary, religious, or cultural-commentary genres.

OO-2: Missing categories: (a) natural-language feedback / revision-suggestion outputs that the deployment requires [elicitation A3, WEB-12]; (b) trait categories or rubric dimensions specific to genres absent from the benchmark (e.g., textual fidelity for Quranic commentary, literary device recognition).

OO-3: INSUFFICIENT DOCUMENTATION on whether CAST trait definitions encode Qatar-specific stylistic norms versus pan-Arab conventions; the regional YAML flags this as gap_id 1.

OO-4: Stakeholder-driven taxonomy redesign would be needed to (i) add a feedback-generation output category and (ii) extend or replace the rubric for non-persuasive genres.

OO-5: Documented taxonomy issues: numeric-only output schema [Q48, Q181, Q147]; CAST genre-specificity [Q118]; narrow REL scale limits granularity [Q91]; no validated mapping from trait scores to actionable feedback.

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WEB-12	AES literature explicitly identifies feedback generation as an overlooked area (https://link.springer.com/article/10.1007/s10462-024-11017-5)

Information Gaps

- Whether trait-score outputs can be reliably converted to feedback through downstream generation models
- Whether CAST trait definitions encode Qatar-specific norms

Requires Expert Verification

- Pedagogical experts on whether the CAST trait decomposition supports formative feedback at all

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Remediation

Gap 1: Output schema is restricted to numeric trait scoring with no feedback-generation category; CAST rubric is genre-specific to persuasive/argumentative writing.

Recommendation: Redesign the output ontology to include a feedback-generation task and extend or replace CAST with rubrics validated for literary analysis, religious commentary, and cultural commentary genres. Engage Qatari and other Arab curriculum specialists to define rubric categories.